

12.8

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Question

Given the wavelets, $a = \{3, -2\}$ and $b = \{1, -2\}$,
The cross-correlation, ϕ_{ab} , is given by

- (a) $\{-6, 7, -2\}$
- (b) $\{-6, 10, -12\}$
- (c) $\{-4, -11, -6\}$
- (d) $\{-6, 11, -4\}$

Let

$$a = (a_0, a_1, \dots, a_{M-1}), \quad b = (b_0, b_1, \dots, b_{N-1}).$$

We usually list $\ell = L_{\min}, \dots, L_{\max}$ with $L_{\min} = -(N-1)$, $L_{\max} = M-1$.

$$\begin{pmatrix} \phi_{ab}[L_{\min}] \\ \phi_{ab}[L_{\min} + 1] \\ \vdots \\ \phi_{ab}[L_{\max}] \end{pmatrix} = T(b) \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_{M-1} \end{pmatrix}.$$

Solution

Given

$$a = \begin{pmatrix} 3 \\ -2 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ -2 \end{pmatrix},$$

we compute the cross-correlation $\phi_{ab}[\ell] = \sum_n a_n b_{n+\ell}$ for $\ell = -1, 0, 1$.

$$\mathbf{T} = \begin{pmatrix} -2 & 0 \\ 1 & -2 \\ 0 & 1 \end{pmatrix}, \quad a = \begin{pmatrix} 3 \\ -2 \end{pmatrix}.$$

Solution

Multiply to get the vector of correlations (ordered by $\ell = -1, 0, 1$):

$$\phi_{ab} = \mathbf{T} \mathbf{a} = \begin{pmatrix} -6 \\ 7 \\ -2 \end{pmatrix}$$

so the correlation sequence is

$$\{-6, 7, -2\}.$$

which is option (a).